Test Series No.

1.11

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 17/03/2026

ANSWER KEY

1.	(1)	46.	(2)	91.	(3)	136.	(1)
2.	(2)	47.	(3)	92.	(4)	137.	(2)
3.	(3)	48.	(1)	93.	(2)	138.	(2)
4.	(1)	49.	(3)	94.	(1)	139.	(1)
5.	(2)	50.	(2)	95.	(2)	140.	(3)
6.	(4)	51.	(2)	96.	(4)	141.	(3)
7.	(4)	52.	(3)	97.	(3)	142.	(3)
8.	(1)	53.	(3)	98.	(1)	143.	(4)
9.	(1)	54.	(1)	99.	(3)	144.	(4)
10.	(4)	55.	(2)	100.	(4)	145.	(1)
11.	(1)	56.	(1)	101.	(2)	146.	(1)
12.	(4)	57.	(1)	102.	(1)	147.	(4)
13.	(2)	58.	(3)	103.	(3)	148.	(1)
14.	(1)	59.	(4)	104.	(2)	149.	(3)
15.	(4)	60.	(1)	105.	(4)	150.	(1)
16.	(1)	61.	(2)	106.	(2)	151.	(1)
17.	(3)	62.	(3)	107.	(3)	152.	(3)
18.	(1)	63.	(2)	108.	(1)	153.	(2)
19.	(2)	64.	(3)	109.	(1)	154.	(4)
20.	(2)	65.	(2)	110.	(4)	155.	(2)
21.	(1)	66.	(4)	111.	(2)	156.	(2)
22.	(2)	67.	(4)	112.	(4)	157.	(3)
23.	(1)	68.	(4)	113.	(3)	158.	(2)
24.	(1)	69.	(2)	114.	(4)	159.	(2)
25.	(1)	70.	(4)	115.	(1)	160.	(4)
26.	(1)	71.	(4)	116.	(2)	161.	(1)
27.	(3)	72.	(1)	117.	(3)	162.	(4)
28.	(1)	73.	(4)	118.	(1)	163.	(3)
29.	(3)	74.	(2)	119.	(4)	164.	(4)
30.	(2)	75.	(3)	120.	(2)	165.	(3)
31.	(2)	76.	(2)	121.	(1)	166.	(3)
32.	(2)	77.	(3)	122.	(3)	167.	(4)
33.	(1)	78.	(3)	123.	(4)	168.	(4)
34.	(1)	79.	(3)	124.	(2)	169.	(1)
35.	(3)	80.	(2)	125.	(1)	170.	(4)
36.	(2)	81.	(4)	126.	(3)	171.	(2)
37.	(3)	82.	(2)	127.	(2)	172.	(3)
38.	(1)	83.	(4)	128.	(4)	173.	(3)
39.	(2)	84.	(1)	129.	(1)	174.	(3)
40.	(4)	85.	(4)	130.	(3)	175.	(2)
41.	(1)	86.	(2)	131.	(2)	176.	(2)
42.	(1)	87.	(1)	132.	(4)	177.	(2)
43.	(1)	88.	(4)	133.	(4)	178.	(4)
44.	(4)	89.	(2)	134.	(1)	179.	(4)
45.	(4)	90.	(3)	135.	(3)	180.	(3)

[Contd...

Hints & Solutions

1. (1)

Using the equation of motion: $S=\left(\frac{u+v}{2}\right)t$

$135=\left(\frac{10+20}{2}\right)t$

$135=15t \Rightarrow t=\frac{135}{15}=9$ s.

(NCERT Ed. 2025-26, 11th, Page No. 20)

2. (2)

Force per unit length $f=\frac{\mu_0 I_1 I_2}{2\pi d}$.

$F \propto \frac{I_1 I_2}{d}$.

New Force $F' \propto \frac{(2I_1)I_2}{3d}=\frac{2}{3}\frac{I_1 I_2}{d}$.

$F'=\frac{2}{3}F$.

(NCERT Ed. 2025-26, 12th, Page No. 123)

3. (3)

Adiabatic expansion: $Q=0$, work is done by gas ($W>0$).

From $\Delta U=Q-W$, $\Delta U=-W$ (decreases).

Hence Temp decreases. Ideal gas $PV=nRT$. Since T decreases, PV must decrease. Both statements are correct.

(NCERT Ed. 2025-26, 11th, Page No. 235)

4. (1)

Wavelength in vacuum

$\lambda_0=c/f=\left(3\times10^8\right)/\left(3\times10^6\right)=100$ m.

Refractive index $n=\sqrt{\epsilon_r}=\sqrt{4}=2$.

Wavelength in medium

$\lambda_m=\lambda_0/n=100/2=50$ m

Change in wavelength $\Delta\lambda=100-50=50$ m.

(NCERT Ed. 2025-26, 12th, Page No. 207)

5. (2)

Fringe width $\beta=\frac{\lambda D}{d}$.

New $\beta'=\frac{\lambda(2D)}{(d/2)}=4\frac{\lambda D}{d}=4\beta$.

(NCERT Ed. 2025-26, 12th, Page No. 266)

6. (4)

Gravitational field $|\vec{g}|=\frac{F}{m}$.

$m=60$ g $=0.06$ kg.

$|\vec{g}|=\frac{3.0}{0.06}=\frac{300}{6}=50$ N/kg

(NCERT Ed. 2025-26, 11th, Page No. 147)

7. (4)

When capacitor is fully charged it blocks dc current

$I=\frac{10}{4}=2.5$ A

$Q=CV=10\times2.5=25\mu$C

(NCERT Ed. 2025-26, 12th, Page No. 74)

8. (1)

$U=n_1\frac{f_1}{2}RT+n_2\frac{f_2}{2}RT$

O_2 (Diatomic): $f=5$

Ar (Monoatomic): $f=3$

$U=2\left(\frac{5}{2}RT\right)+4\left(\frac{3}{2}RT\right)=5RT+6RT=11RT$

(NCERT Ed. 2025-26, 11th, Page No. 249)

9. (1)

$R_{eq}=\frac{R_1R_2}{R_1+R_2}=\frac{16}{8}=2\Omega$

Error in parallel: $\frac{\Delta R_{eq}}{R_{eq}^2}=\frac{\Delta R_1}{R_1^2}+\frac{\Delta R_2}{R_2^2}$

$\frac{\Delta R_{eq}}{4}=\frac{0.8}{16}+\frac{0.4}{16}=\frac{1.2}{16}$

$\Delta R_{eq}=4\times\frac{1.2}{16}=0.3$.

Result: $(2\pm0.3)\Omega$.

(NCERT Ed. 2025-26, 11th, Page No. 9)

10. (4)

$|\varepsilon_{avg}|=\left|\frac{\Delta\phi}{\Delta t}\right|=\left|\frac{\phi(t_2)-\phi(t_1)}{t_2-t_1}\right|$

$\phi(2)=5(2)^2+3(2)+16$

$\phi(2)=20+6+16=42$ Wb

$\phi(6)=5(6)^2+3(6)+16$

[Contd...

$\phi(6) = 180 + 18 + 16 = 214$ Wb

$\left|\varepsilon_{avg}\right| = \frac{214 - 42}{4} = 43$ volt

(NCERT Ed. 2025-26, 12th, Page No. 158)

11. (1)

Current $I = \frac{e}{T} = \frac{e}{\left(\frac{2\pi r}{v}\right)} = \frac{ev}{2\pi r}$

$B = \frac{\mu_0 I}{2r} = \frac{\mu_0 (ev/2\pi r)}{2r} = \frac{\mu_0 ev}{4\pi r^2}$.

$r^2 \propto \frac{v}{B} \Rightarrow r \propto \sqrt{\frac{v}{B}}$.

(NCERT Ed. 2025-26, 12th, Page No. 116)

12. (4)

In β^+ decay (positron emission), a neutrino (ν) is emitted.

(NCERT Ed. 2025-26, 12th, Page No. 314)

13. (2)

$\Delta U = U_f - U_i = \frac{-GMm}{R+h} - \frac{-GMm}{R}$

Here $h = R$

$\Delta U = \frac{-GMm}{2R} + \frac{GMm}{R} = \frac{GMm}{2R}$

Since $\frac{GM}{R^2} = g$,

$\frac{GM}{R} = gR$

$\Delta U = \frac{1}{2}mgR$

(NCERT Ed. 2025-26, 11th, Page No. 152)

14. (1)

Acceleration $a = \frac{F}{M}$

Mass of the remaining portion (having length $L - x$) is

$m' = \frac{M}{L}(L - x)$.

Tension pulls this remaining mass.

$T = m'a = \frac{M}{L}(L - x) \times \frac{F}{M} = F\frac{L - x}{L} = F\left(1 - \frac{x}{L}\right)$.

(NCERT Ed. 2025-26, 11th, Page No. 51)

15. (4)

Power factor $= \frac{r}{z}$

$= \frac{15}{\sqrt{15^2 + (25 - 5)^2}} = 0.6$

(NCERT Ed. 2025-26, 12th, Page No. 191)

16. (1)

Statement I is Correct. Statement II is Incorrect Concave mirror forms virtual, erect, magnified image when object is between Pole and Focus.

(NCERT Ed. 2025-26, 12th, Page No. 226)

17. (3)

Electrical Capacitance (C):

$U = \frac{1}{2}\frac{Q^2}{C} \Rightarrow C = \frac{Q^2}{2U}$

Dimensions:

$\frac{[AT]^2}{[ML^2T^{-2}]} = [M^{-1}L^{-2}T^4A^2]$ Matches (II).

Magnetic Permeability (μ_0):

Force per unit length between parallel wires

$f = \frac{\mu_0 I_1 I_2}{2\pi d}$

$\mu_0 = \frac{F \cdot d}{l \cdot I^2}$ (Dimensionally)

Dimensions:

$\frac{[MLT^{-2}][L]}{[L][A^2]} = [MLT^{-2}A^{-2}]$ Matches (IV).

Planck's Constant (h):

$E = h\nu \Rightarrow h = E/\nu$

Dimensions:

$\frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$ Matches (I).

Young's Modulus (Y):

Formula: $Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{\Delta L/L}$

Dimensions:

$\frac{[MLT^{-2}]/[L^2]}{1} = [ML^{-1}T^{-2}]$ Matches (III).

(NCERT Ed. 2025-26, 11th, Page No. 15)

[Contd...

18. (1)

After first polaroid, intensity $I_1 = \frac{I_0}{2}$.

After second polaroid (Malus Law): $I_2 = I_1\cos^2\theta$.

$$I_2 = \left(\frac{I_0}{2}\right)\cos^2(60°) = \left(\frac{I_0}{2}\right)\left(\frac{1}{2}\right)^2 = \frac{I_0}{8}.$$

(NCERT Ed. 2025-26, 12th, Page No. 271)

19. (2)

For frictionless banked road:

$$\tan\theta = \frac{v^2}{rg}$$

tan 45° = 1.

$1 = v^2/(90\times10)$.

$v^2 = 900 \Rightarrow v = 30$ m/s.

(NCERT Ed. 2025-26, 11th, Page No. 64)

20. (2)

$$U = \frac{1}{2}Y(\text{strain})^2 \times \text{Volume}$$

$$U = \frac{YA(\Delta l)^2}{2L}$$

$$Y = \frac{2UL}{A(\Delta l)^2}$$

$$Y = \frac{2\times0.4\times2}{(4\times10^{-6})\times(2\times10^{-3})^2}$$

$$Y = \frac{1.6}{(4\times10^{-6})\times(4\times10^{-6})}$$

$$Y = \frac{1.6}{16\times10^{-12}}$$

$Y = 0.1\times10^{12} = 1.0\times10^{11}$ N/m^2

(NCERT Ed. 2025-26, 11th, Page No. 174)

21. (1)

Open pipe fundamental $f_o = \frac{v}{2L_o} = 300$

First overtone open = $2f_o = 600$ Hz.

Closed pipe first overtone (3rd harmonic)

$f_c = 3v/4L_c$.

Given $600 = 3(330)/4L_c$.

$$L_c = \frac{990}{2400\text{ m}} = \frac{99}{240\text{ m}} \approx 0.412\text{ m} = 41.2\text{ cm}$$

(NCERT Ed. 2025-26, 11th, Page No. 292)

22. (2)

$$r = \frac{p}{qB}$$

Since p and B are same, $r \propto \frac{1}{q}$

$q_p = e$, $q_\alpha = 2e$.

$$\frac{r_p}{r_\alpha} = \frac{q_\alpha}{q_p} = \frac{2e}{e} = 2:1.$$

(NCERT Ed. 2025-26, 12th, Page No. 112)

23. (1)

Assertion is True (Displacement is 0).

Reason is False (Distance is $4A$, so average speed is $4A/T \neq 0$).

(NCERT Ed. 2025-26, 11th, Page No. 262)

24. (1)

$\Delta Q = -20$ J (Released).

$\Delta W = -8$ J (Work done ON gas).

First Law: $\Delta Q = \Delta U + \Delta W$

$-20 = (U_f - 30) + (-8)$

$-20 = U_f - 38$.

$U_f = 18$ J

(NCERT Ed. 2025-26, 11th, Page No. 230)

25. (1)

For equilibrium, forces must cancel.

$$\frac{k(9e)q}{x^2} = \frac{k(e)q}{(16-x)^2}.$$

Taking root: $\frac{3}{x} = 1/(16-x)$.

$48 - 3x = x \Rightarrow 4x = 48 \Rightarrow x = 12$ cm.

(NCERT Ed. 2025-26, 12th, Page No. 12)

26. (1)

Conservation of Angular Momentum:

$L = mv_1r_1 = mv_2r_2$.

$$v_2 = v_1\left(\frac{r_1}{r_2}\right).$$

(NCERT Ed. 2025-26, 11th, Page No. 144)

[Contd...

27. (3)

Using $v^2 - u^2 = 2gh$

$(80)^2 - (20)^2 = 2(10)h$

$6400 - 400 = 20h$

$6000 = 20h \Rightarrow h = 300$ m

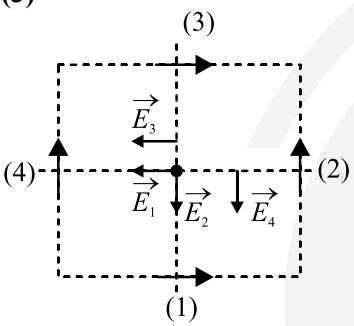
(NCERT Ed. 2025-26, 11th, Page No. 20)

28. (1)

Assertion is correct: Higher temp breaks bonds, creating more carriers (n increases), lowering resistance. Reason is incorrect: Relaxation time (τ) actually decreases with temperature due to more collisions, but the increase in carrier concentration n dominates the effect in semiconductors.

(NCERT Ed. 2025-26, 12th, Page No. 324)

29. (3)



All electric fields will have equal magnitude

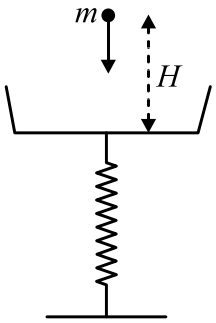
$E_{net} = 2\sqrt{2}E$

$E = \frac{8KP}{L^3}$

$\therefore\ E_{net} = \frac{16\sqrt{2}KP}{L^3}$

(NCERT Ed. 2025-26, 12th, Page No. 16)

30. (2)



Number of particles fallen in time t in nt

So weight of particles reached is $(nmt)g$ and impulsive force given is $mn\sqrt{2gH}$

$\therefore$ net reading of weighing machine is

$mn(gt + \sqrt{2gH})$

(NCERT Ed. 2025-26, 11th, Page No. 53)

31. (2)

Energy corresponding to an emitted spectral line

$\Delta E = E_2 - E_1 \propto Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

In both the cases, for a given transition. $\nu \propto Z^2$

$\frac{\nu_{He}}{\nu_H} = \left(\frac{Z_{He}}{Z_H}\right)^2 = \left(\frac{2}{1}\right)^2 = 4;$

$\nu_{He} = 4\nu_H$

(NCERT Ed. 2025-26, 12th, Page No. 299)

32. (2)

Let x be the number of cells in reversed connection

$84 = 3 \times 32 - 2x \times 3$ or $x = 2$

Hence, the correct answer is option (2).

(NCERT Ed. 2025-26, 12th, Page No. 96)

33. (1)

Initial speed, $v_i = 40$ ms^{-1}

Final speed, $v_f = 20$ ms^{-1}

$\therefore\ KE_i = \frac{1}{2} \times 3 \times 40^2 = 2400$ J

$KE_f = \frac{1}{2} \times 3 \times 20^2 = 600$ J

$\Delta KE = 600 - 2400 = -1800$ J

$\therefore$ From work – energy theorem, work done by all forces is $W_{\text{net}} = \Delta KE = -1800$ J

(NCERT Ed. 2025-26, 11th, Page No. 76)

34. (1)

According to theorem of parallel axes,

$I = \frac{2}{5}M\left(\frac{R}{2}\right)^2 + M(2R)^2 + \frac{2}{5}M\left(\frac{R}{2}\right)^2$

$= 4MR^2 + \frac{1}{5}MR^2 = \frac{21}{5}MR^2$

(NCERT Ed. 2025-26, 11th, Page No. 115)

35. (3)

For a rotating conductor of an arbitrary shape in uniform magnetic field $\vec{B}$, induced emf is

$\varepsilon_{ba} = \frac{1}{2}B\omega L^2$

$\varepsilon_{PQ} = \frac{1}{2}B\omega(2l)^2 = 2B\omega l^2 = 100$ V

$\varepsilon_{PM} = \frac{1}{2}B\omega l^2 = \frac{1}{4}\varepsilon_{PQ} = 25$ V

$(\varepsilon_P - \varepsilon_Q) - (\varepsilon_P - \varepsilon_M) = 100 - 25$

$\therefore\ \varepsilon_{MQ} = 75$ V

(NCERT Ed. 2025-26, 12th, Page No. 164)

[Contd...

36. (2)

Let the final temperature be T

Heat required by ice $= mL + m \times s \times (T - 0)$

$= 10 \times 80 + 10 \times 1 \times T$

Heat lost by tumbler = $55 \times (40 - T)$

By using law of calorimetry,

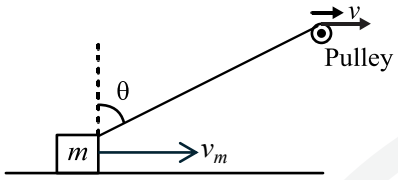
heat gained = heat lost

$800 + 10T = 55 \times (40 - T)$

$\Rightarrow T = 21.54°C \approx 22°C$

(NCERT Ed. 2025-26, 11th, Page No. 209)

37. (3)



From constraint relation,

$v_m \sin\theta = v$

$\Rightarrow v_m = \dfrac{v}{\sin\theta}$

(NCERT Ed. 2025-26, 11th, Page No. 29)

38. (1)

Kinetic energy of thermal neutron $E = \dfrac{3}{2}kT$.

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2m(3/2kT)}} = \frac{h}{\sqrt{3mkT}}$$

(NCERT Ed. 2025-26, 12th, Page No. 285)

39. (2)

$n_e.n_h = n_i^2$

$\Rightarrow n_e = \dfrac{(7.07\times10^{15})^2}{5\times10^{22}} \approx 10^9 \text{ m}^{-3}$

(NCERT Ed. 2025-26, 12th, Page No. 332)

40. (4)

$m = \dfrac{f_0}{f_e} = 5$

$f_0 + f_e = 36$

f_e = 6 cm

f_0 = 30 cm

(NCERT Ed. 2025-26, 12th, Page No. 244)

41. (1)

At the origin $V = 0 \Rightarrow$ option 3 and 4 are incorrect

When $-a < x < 0$ potential should be positive as charge is positive and when $0 < x < a$ potential should be negative as charge is negative.

$\therefore$ option 1 is correct.

(NCERT Ed. 2025-26, 12th, Page No. 49)

42. (1)

$\theta = 2t^3 - 6t^2$

$\dfrac{d\theta}{dt} = \omega = 6t^2 - 12t$

$\alpha = \dfrac{d\omega}{dt} = 12t - 12$

At t = 1s, $\alpha = 0$ $\therefore$ $\tau = 0$

(NCERT Ed. 2025-26, 11th, Page No. 106)

43. (1)

$\phi = \vec{E}.\vec{A}$

$\phi = 4\hat{i}.(2\hat{i} + 3\hat{j})$

ϕ = 8 V-m

(NCERT Ed. 2025-26, 12th, Page No. 22)

44. (4)

Using equation of continuity

$Av = A_1v_1 + A_2v_2 + A_3v_3$

$0.3(3) = 0.2(2) + 0.3(1) + 0.1(v)$

$0.9 - 0.4 - 0.3 = 0.1(v)$

$v = \dfrac{0.2}{0.1} = 2 \text{ m/s}$

(NCERT Ed. 2025-26, 11th, Page No. 187)

45. (4)

$F \propto \dfrac{\mu_0}{4\pi}\dfrac{M_1M_2}{d^4}$

So, $F \propto \dfrac{1}{d^4}$

(NCERT Ed. 2025-26, 12th, Page No. 140)

[Contd...

46. (2)

Due to diagonal relationship, aluminium resembles beryllium in electronegativity because comparable charge density and polarising power occur in their compounds.

(NCERT Ed. 2025-26 11th Part-I Page No. 93)

47. (3)

Negative resonance withdrawers have a carbonyl attached; –COOH pulls electron density by –R, while $–NH_2$, –OH, –OR donate to ring.

(NCERT Ed. 2025-26 11th Part-II Page No. 276)

48. (1)

$K = \frac{K_2}{K_1^2}$, $= \frac{10^{100}}{10^{80}}$ giving 10^{20} as equilibrium constant for the reaction.

(NCERT Ed. 2025-26 11th Part-I Page No. 178)

49. (3)

Chromium charges from +6 in chromate to +3, a decrease of three; permanganate to Mn^{2+} changes five and Cl_2 to ClO_3^- changes 5.

(NCERT Ed. 2025-26 11th Part-II Page No. 239)

50. (2)

Ferrocyanide forms Prussian blue only with Fe^{3+}, producing ferric ferrocyanide complex; Fe^{2+} gives pale precipitate instead in this test solution.

51. (2)

Moles NaCl equal 2.925/58.5; volume 0.25 L, so molarity equals 0.20 M.

(NCERT Ed. 2025-26 12th Part-I Page No. 04)

52. (3)

For $BaCl_2$, i = 3 so $\pi = 3 \times 0.02$ RT; $\pi_{xy} = 3\pi_{BaCl_2}$ and i = 2, hence M = 0.09 since osmotic pressure is proportional to i times molarity here.

(NCERT Ed. 2025-26 12th Part-I Page No. 22)

53. (3)

Half-life equals 0.693/k = 300 s ; 25% remaining needs two half-lives, 12.5% needs three; units are s^{-1}, so four statements hold.

(NCERT Ed. 2025-26 12th Part-I Page No. 69)

54. (1)

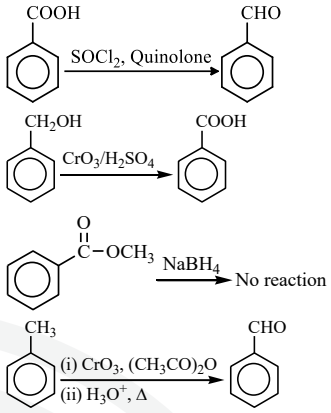
TeO_2 acts as an oxidising agent, while H_2Te is acidic; thus pair is oxidising and acidic as required.

55. (2)

Mn shows +7 in permanganate, Ru and Os show +8 in oxides, and Cr(+6) is oxidising;
Sc(+4) false statement there

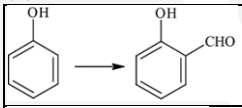
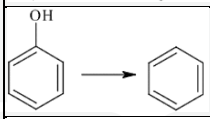
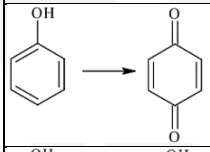
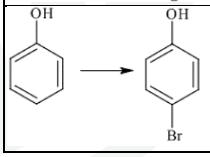
(NCERT Ed. 2025-26 12th Part-I Page No. 96)

56. (1)



(NCERT Ed. 2025-26 12th Part-II Page No. 232)

57. (1)

	$CHCl_3$/NaOH
	Zn
	$Na_2Cr_2O_7/H_2SO_4$
	Br_2 in CS_2

(NCERT Ed. 2025-26 12th Part-II Page No. 213)

58. (3)

2,4-DNP reacts with carbonyl groups forming orange hydrazones; thus, aldehydes and ketones respond, not others listed in this confirmatory test.

(NCERT Ed. 2025-26 12th Part-II Page No. 238)

59. (4)

Aspartic acid has lowest pH due extra –COOH; basic Arg highest; Gly neutral, Lys basic intermediate in between these values.

(NCERT Ed. 2025-26 12th Part-II Page No. 291)

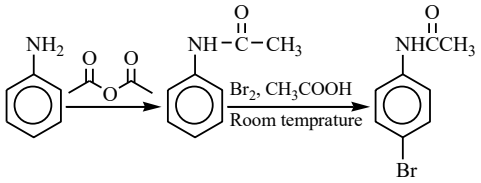
60. (1)

Nitrating mixture protonates aniline to anilinium ion, a meta director, so m-product appears with o,p products during usual cold nitration.

(NCERT Ed. 2025-26 12th Part-II Page No. 273)

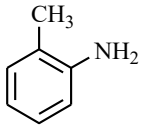
[Contd...

61. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 272)

62. (3)



In this NH_2 is more ring activating than CH_3

(NCERT Ed. 2025-26 11th Part-II Page No. 276)

63. (2)

Stronger ligand gives larger Δ_0, absorbing higher-energy shorter-wavelength light; violet strongest, then green, orange weakest since energy equals hc/λ always.

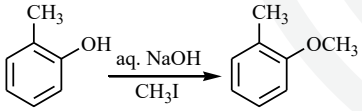
(NCERT Ed. 2025-26 12th Part-I Page No. 132)

64. (3)

Across actinides, ionic radius decreases with increasing atomic number due to actinoid contraction; therefore Cm^{3+} is smaller than U^{3+} in solution.

(NCERT Ed. 2025-26 12th Part-I Page No. 112)

65. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 215)

66. (4)

On heating $KMnO_4$ it disproportionates to K_2MnO_4, MnO_2 and O_2; manganate(VI) is green and paramagnetic because Mn +6 has d^1.

(NCERT Ed. 2025-26 12th Part-I Page No. 106)

67. (4)

In alkaline medium Ni^{2+} forms chelate $Ni(DMG)_2$ giving characteristic scarlet precipitate; other reagents do not produce it selectively under conditions.

68. (4)

Vitamin B_{12} is water-soluble yet stored in liver, so it accumulates; B_1, B_2, B_6 are rapidly excreted in urine daily.

(NCERT Ed. 2025-26 12th Part-II Page No. 296)

69. (2)

Cryolite is Na_3AlF_6; writing Na_2AlF_6 gives wrong stoichiometry, so that pair is incorrect among options others match known mineral formulas.

70. (4)

Interstitial compounds retain metallic conductivity, hardness and high melting points, but are generally chemically inert, not poor conductors at all.

(NCERT Ed. 2025-26 12th Part-II Page No. 104)

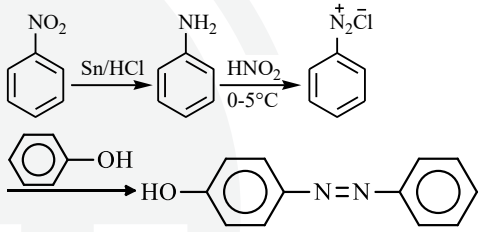
71. (4)

RNA polymerase reads the template DNA strand 3' → 5', but synthesizes RNA 5' → 3'

(so, Statement-I is incorrect, Statement-II is correct).

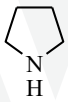
(NCERT Ed. 2025-26 12th Part-II Page No. 299)

72. (1)



(NCERT Ed. 2025-26 12th Part-II Page No. 276)

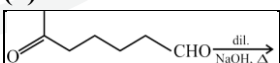
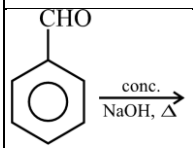
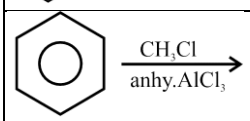
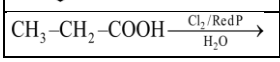
73. (4)



is most basic sp^3 hybridised 2°.

(NCERT Ed. 2025-26 12th Part-II Page No. 276)

74. (2)

	Intramolecular aldol condensation
	Cannizzaro reaction
	Friedel Craft's alkylation
	HVZ reaction

(NCERT Ed. 2025-26 12th Part-II Page No. 241)

75. (3)

Dilute sulphuric acid supplies H^+, suppressing Fe^{2+} hydrolysis and oxidation; nitric oxidizes, hydrochloric complexes, concentrated is unnecessary in Mohr solution.

[Contd...

76. (2)

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

given 1 : 1 $CH_4 : CO_2$; 8.8 g CO_2 is 0.2 mol, so x = 20.

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

77. (3)

For hydrogen-like ions,

- $KE \propto \frac{Z^2}{n^2}$
- $V \propto \frac{Z}{n}$ $Vn \propto Z$
- $f \propto \frac{Z^2}{n^3}$

(NCERT Ed. 2025-26 11th Part-I Page No. 48)

78. (3)

Half-filled subshells have symmetric electron distribution and maximum exchange energy; reduced pairing lowers repulsion, giving extra stability in many atoms

(NCERT Ed. 2025-26 11th Part-I Page No. 65)

79. (3)

Internal energy, enthalpy and volume depend only on state, whereas heat depends on path; hence three state variables among listed.

(NCERT Ed. 2025-26 11th Part-I Page No. 138)

80. (2)

H_2O is bent with two polar O–H bonds adding; CO_2 and BF_3 cancel, NF_3 has small net moment overall here.

81. (4)

μ: $CH_3Cl > CH_3F > CH_3Br > CH_3I$

(NCERT Ed. 2025-26 12th Part-II Page No. 164)

82. (2)

$$\Delta S_{surr} = -\frac{\Delta_r H^o}{T} = -\frac{(-1648\times10^3)}{298} = 5530\ JK^{-1}\ mol^{-1}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 160)

83. (4)

- Mercury cell: (Cathode)

  $Zn(Hg) + 2OH^- \rightarrow ZnO(s) + H_2O + 2e^-$
- Lead storage battery: (Cathode)

  $PbO_2 + SO_4^{2-} + 4H^+ + 2e^- \rightarrow PbSO_4 + 2H_2O$
- Dry cell: (Cathode)

  $MnO_2 + NH_4^+ + e^- \rightarrow MnO(OH) + NH_3$
- Fuel cell: (Cathode)

  $O_2(g) + 2H_2O + 4e^- \rightarrow 4OH^-$

(NCERT Ed. 2025-26 12th Part-I Page No. 54)

84. (1)

During group detection, 'Group V' cation Ba^{2+} is precipitated as carbonates using the group reagent $(NH_4)_2CO_3$ in presence of NH_4OH.

85. (4)

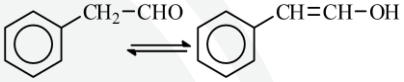
s-block and p-block elements are collectively known as representative elements.

Group-16 elements are known as chalcogens.

Atomic radius of noble gases is expressed as van der Waal's radius.

86. (2)

Carbonyl compounds containing α-H atom show tautomerism



(NCERT Ed. 2025-26 11th Part-II Page No. 270)

87. (1)

$$k_f = \frac{R \times M_1 \times T_f^2}{1000 \times \Delta_{fus}H}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 19)

88. (4)

$Fe^{2+} + 2e^- \rightarrow Fe(s)$; $E_1^o = -0.44\ V, n_1 = 2$

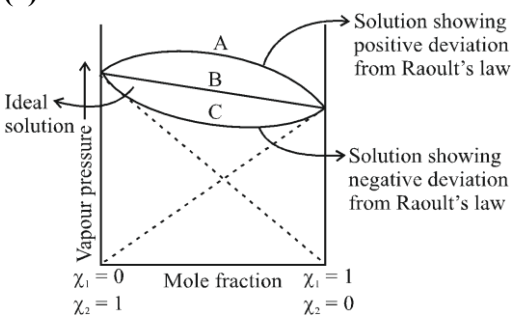
$Fe^{3+} + e^- \rightarrow Fe^{2+}$; $E_2^o = 0.77\ V, n_2 = 1$

$Fe^{3+} + 3e^- \rightarrow Fe(s)$; $E_3^o = ?, n_3 = 3$

$$E_3 = \frac{n_1E_1 + n_2E_2}{n_3} = \frac{2(-0.44) + 1(0.77)}{3} = -0.03\ V$$

(NCERT Ed. 2025-26 12th Part-I Page No. 39)

89. (2)



90. (3)

Compound in which four different groups are attached to the tetrahedral carbon is chiral.

(NCERT Ed. 2025-26 12th Part-I Page No. 14)

[Contd...

91. (3)
The cells that do not divide further exit G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle. Cells in this stage remain metabolically active but no longer proliferate unless called on to do so depending on the requirement of the organism.
(NCERT Ed. 2025-26, 11th, Page No. 122)

92. (4)
A nucleotide has three components – a nitrogenous base, a pentose sugar (ribose in case of RNA, and deoxyribose for DNA), and a phosphate group. When a phosphate group is linked to OH of 5' C of a nucleoside through phosphoester linkage, a corresponding nucleotide (or deoxynucleotide depending upon the type of sugar present) is formed.
(NCERT Ed. 2025-26, 12th, Page No. 80)

93. (2)
A given species may occupy more than one trophic level in the same ecosystem at the same time; for example, a sparrow is a primary consumer when it eats seeds, fruits, peas, and a secondary consumer when it eats insects and worms. A sparrow eating any seed is no less a predator.
(NCERT Ed. 2025-26, 12th, Page No. 213)

94. (1)
In a dicot root, all tissues on the innerside of the endodermis such as pericycle, vascular bundles and pith constitute the stele.
(NCERT Ed. 2025-26, 11th, Page No. 74)

95. (2)
An example of microbial biocontrol agents that can be introduced in order to control butterfly caterpillars is the bacteria *Bacillus thuringiensis* (often written as *Bt*). These are available in sachets as dried spores which are mixed with water and sprayed onto vulnerable plants such as brassicas and fruit trees, where these are eaten by the insect larvae.
(NCERT Ed. 2025-26, 12th, Page No. 157)

96. (4)

Red alga	*Polysiphonia*
Green alga	*Chara*
Brown alga	*Sargassum*
Blue-green alga	*Nostoc*

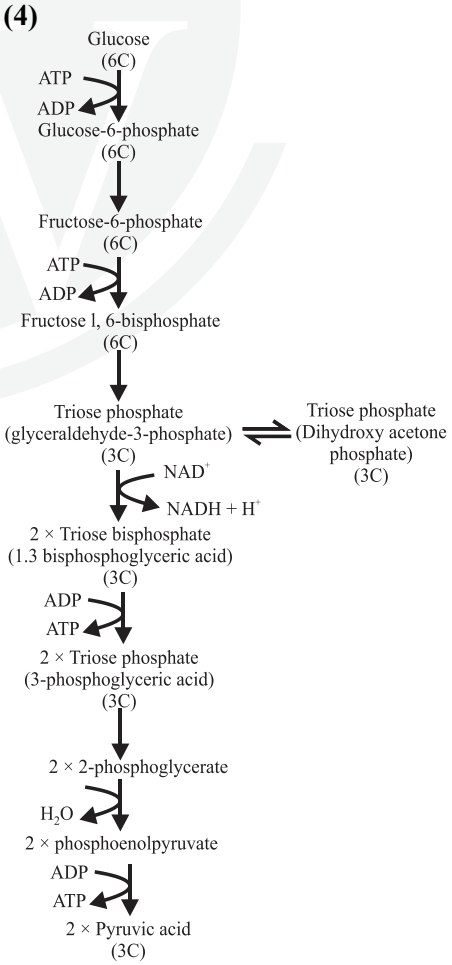
(NCERT Ed. 2025-26, 11th, Page No. 26-28)

97. (3)
Some examples of recent extinctions include the dodo (Mauritius), quagga (Africa), thylacine (Australia), Steller's Sea Cow (Russia) and three subspecies (Bali, Javan, Caspian) of tiger.
(NCERT Ed. 2025-26, 12th, Page No. 221)

98. (1)
In marginal placentation the placenta forms a ridge along the ventral suture of the ovary and the ovules are borne on this ridge forming two rows, as in pea. When the placenta is axial and the ovules are attached to it in a multilocular ovary, the placentation is said to be axile, as in china rose, tomato and lemon. In parietal placentation, the ovules develop on the inner wall of the ovary or on peripheral part...e.g., mustard and *Argemone*. When the ovules are borne on central axis and septa are absent, as in *Dianthus* and *Primrose* the placentation is called free central. In basal placentation, the placenta develops at the base of ovary and a single ovule is attached to it, as in sunflower, marigold.
(NCERT Ed. 2025-26, 11th, Page No. 65)

99. (3)
Some examples of water pollinated plants are *Vallisneria* and *Hydrilla* which grow in fresh water and several marine sea-grasses such as *Zostera*. Not all aquatic plants use water for pollination. In a majority of aquatic plants such as water hyacinth (*Eicchornia*) and water lily, the flowers emerge above the level of water and are pollinated by insects or wind as in most of the land plants.
(NCERT Ed. 2025-26, 12th, Page No. 13)

100. (4)



Steps of glycolysis
(NCERT Ed. 2025-26, 11th, Page No. 156)

101. (2)

Sickle-cell anaemia is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule. Phenyloketonuria is caused by mutation in the gene that codes for the enzyme phenyl alanine hydroxylase. α Thalassemia is controlled by two closely linked genes HBA1 and HBA2 on chromosome 16 of each parent and it is observed due to mutation or deletion of one or more of the four genes. The more genes affected, the less alpha globin molecules produced. While β Thalassemia is controlled by a single gene HBB on chromosome 11 of each parent and occurs due to mutation of one or both the genes.





Micrograph of the red blood cells and the amino acid composition of the relevant portion of β-chain of haemoglobin: (a) From a normal individual; (b) From an individual with sickle-cell anaemia

(NCERT Ed. 2025-26, 12th, Page No. 75)

102. (1)

In many species of fig trees, there is a tight one-to-one relationship with the pollinator species of wasp. It means that a given fig species can be pollinated only by its 'partner' wasp species and no other species. The female wasp uses the fruit not only as an oviposition (egg-laying) site but uses the developing seeds within the fruit for nourishing its larvae. The wasp pollinates the fig inflorescence while searching for suitable egg-laying sites. In return for the favour of pollination the fig offers the wasp some of its developing seeds, as food for the developing wasp larvae.

(NCERT Ed. 2025-26, 12th, Page No. 202)

103. (3)

The TCA cycle (Krebs' cycle) occurs in the matrix of mitochondria.

(NCERT Ed. 2025-26, 11th, Page No. 158)

104. (2)

The Mycoplasma are organisms that completely lack a cell wall. They are the smallest living cells known and can survive without oxygen.

(NCERT Ed. 2025-26, 11th, Page No. 14)

105. (4)

In most higher plants, the growing apical bud inhibits the growth of the lateral (axillary) buds, a phenomenon called apical dominance. It is due to auxins. Cytokinins help overcome the apical dominance.

(NCERT Ed. 2025-26, 11th, Page No. 175, 177)

106. (2)

Katherine Esau reported in her early publications that the curly top virus spreads through a plant via the food-conducting or phloem tissue.

(NCERT Ed. 2025-26, 11th, Page No. 56)

107. (3)

Though the C_4 plants have the C_4 oxaloacetic acid as the first CO_2 fixation product they use the C_3 pathway or the Calvin cycle as the main biosynthetic pathway. In C_4 plants photorespiration does not occur. This is because they have a mechanism that increases the concentration of CO_2 at the enzyme (RuBisCO) site. In C_{33} plants some O_2 does bind to RuBisCO, and hence CO_2 fixation is decreased. Here the RuBP instead of being converted to 2 molecules of PGA binds with O_2 to form one molecule of phosphoglycerate and phosphoglycolate (2 Carbon) in a pathway called photorespiration.

(NCERT Ed. 2025-26, 11th, Page No. 145, 147)

108. (1)

Housefly : Family - Muscidae, Order - Diptera, Class - Insecta, Phylum - Arthropoda.

(NCERT Ed. 2025-26, 11th, Page No. 8)

109. (1)

Lactose is the substrate for the enzyme beta-galactosidase and it regulates switching on and off of the operon. Hence, it is termed as inducer. In the presence of an inducer, such as lactose or allolactose, the repressor is inactivated by interaction with the inducer. This allows RNA polymerase access to the promoter and transcription proceeds.

(NCERT Ed. 2025-26, 12th, Page No. 101)

110. (4)

The chloroplasts contain 70S ribosomes (50S and 30S subunits). The eukaryotic ribosomes are 80S having 60S and 40S subunits while that of 70S ribosomes are 50S and 30S.

(NCERT Ed. 2025-26, 11th, Page No. 98)

111. (2)

By the process of leaching, water-soluble inorganic nutrients go down into the soil horizon and get precipitated as unavailable salts.

(NCERT Ed. 2025-26, 12th, Page No. 207-208)

112. (4)

In grasses, the guard cells are dumb-bell shaped.

(NCERT Ed. 2025-26, 11th, Page No. 72)

[Contd...

113. (3)
Inheritance of flower colour and seed colour in garden pea follows complete dominance. In these both, at F_2, the phenotypic ratio is 3:1 and the genotypic ratio is 1:2:1. Inheritance of flower colour in dogflower is a classic example of incomplete dominance. In it, both the phenotypic ratio and the genotypic ratio is 1:2:1 at F_2. Skin colour in humans is an example of polygenic inheritance (controlled by multiple genes), which does not follow a simple monohybrid ratio.
(NCERT Ed. 2025-26, 12th, Page No. 60)

114. (4)
In deuteromycetes, the asexual spores are conidia while aplanospores are asexual spores produced by members of phycomycetes.
(NCERT Ed. 2025-26, 11th, Page No. 18)

115. (1)
(C) One turn of DNA helix: In B-DNA, one complete turn of the helix contains 10 base pairs (bp).
(B) Typical nucleosome: A typical nucleosome consists of DNA wrapped around a histone octamer, which contains approximately 200 bp of DNA.
(E) DNA of ϕ×174 bacteriophage: This bacteriophage has a genome consisting of 5,386 nucleotides.
(A) DNA content in *E. coli*: *Escherichia coli* has approximately 4.6×10^6 bp (4.6 million base pairs).
(D) Diploid content of human DNA: The diploid content is 6.6×10^9 bp (6.6 billion base pairs).
(NCERT Ed. 2025-26, 12th, Page No. 80-83)

116. (2)
The stems in gymnosperms are unbranched (*Cycas*) or branched (*Pinus, Cedrus*).
(NCERT Ed. 2025-26, 11th, Page No. 32)

117. (3)
Production of cleistogamous flowers promotes self-pollination.
(NCERT Ed. 2025-26, 12th, Page No. 15)

118. (1)
To make one molecule of glucose, 6 turns of the Calvin cycle are required.
(NCERT Ed. 2025-26, 11th, Page No. 145)

119. (4)
Societies which indulged in anthropocentric view of biology could register limited progress in biological knowledge.
(NCERT Ed. 2025-26, 11th, Page No. 1)

120. (2)
A very significant contribution of mitosis is cell repair. Mitotic divisions in the meristematic tissues – the apical and the lateral cambium, result in a continuous growth of plants throughout their life.
(NCERT Ed. 2025-26, 11th, Page No. 125)

121. (1)
Species composition and stratification are the two main structural features of an ecosystem. Productivity, decomposition, energy flow, and nutrient cycling are the four important components of an ecosystem.
(NCERT Ed. 2025-26, 12th, Page No. 206)

122. (3)
In grasshoppers, sex determination is of the XO type. Males have only one X chromosome and lack a second sex chromosome while the females possess two X chromosomes. In birds, the sex determination system is ZZ/ZW. The males are homogametic (ZZ) and females are heterogametic, possessing two different sex chromosomes (ZW).
(NCERT Ed. 2025-26, 12th, Page No. 70-71)

123. (4)
Ethylene is used to initiate flowering and for synchronising fruit-set in pineapples. It also induces flowering in mango. It promotes female flowers in cucumbers thereby increasing the yield.
(NCERT Ed. 2025-26, 11th, Page No. 177)

124. (2)
The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake.
(NCERT Ed. 2025-26, 12th, Page No. 223)

125. (1)
In plants like mint and jasmine a slender lateral branch arises from the base of the main axis and after growing aerially for some time arch downwards to touch the ground. In banana, pineapple and *Chrysanthemum*, the lateral branches originate from the basal and underground portion of the main stem, grow horizontally beneath the soil and then come out obliquely upward giving rise to leafy shoots. Stem tendrils which develop from axillary buds, are slender and spirally coiled and help plants to climb such as in gourds (cucumber, pumpkins, watermelon) and grapevines. Some plants of arid regions modify their stems into flattened (*Opuntia*), or fleshy cylindrical (*Euphorbia*) structures. They contain chlorophyll and carry out photosynthesis.
(NCERT Ed. 2025-26, 11th, Page No. 68)

[Contd...

126. (3)

Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*).

(NCERT Ed. 2025-26, 11th, Page No. 88)

127. (2)

Down's Syndrome : The cause of this genetic disorder is the presence of an additional copy of the chromosome number 21 (trisomy of 21).

(NCERT Ed. 2025-26, 12th, Page No. 76)

128. (4)

In 1928, Frederick Griffith, in a series of experiments with *Streptococcus pneumoniae* (bacterium responsible for pneumonia), witnessed a miraculous transformation in the bacteria. It is now proven that DNA replicates semi-conservatively. It was shown first in *Escherichia coli* by Meselson and Stahl. Very similar experiments involving use of radioactive thymidine to detect distribution of newly synthesised DNA in the chromosomes was performed on *Vicia faba* (faba beans) by Taylor and colleagues in 1958. Morgan worked with the tiny fruit flies, *Drosophila melanogaster*.

(NCERT Ed. 2025-26, 12th, Page No. 67, 84, 89, 90)

129. (1)

The flower is epigynous in the members of family Compositae.

(NCERT Ed. 2025-26, 11th, Page No. 63)

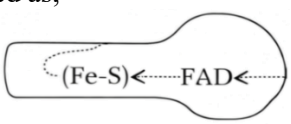
130. (3)

The density of a population in a given habitat during a given period, fluctuates due to changes in four basic processes, two of which (natality and immigration) contribute to an increase in population density and two (mortality and emigration) to a decrease. Under normal conditions, births and deaths are the most important factors influencing population density, the other two factors assuming importance only under special conditions. For instance, if a new habitat is just being colonised, immigration may contribute more significantly to population growth than birth rates.

(NCERT Ed. 2025-26, 12th, Page No. 193)

131. (2)

In the mitochondrial ETS, complex II is represented as;



(NCERT Ed. 2025-26, 11th, Page No. 160)

132. (4)

In a transverse section, a typical microsporangium appears near circular in outline. It is generally surrounded by four wall layers – the epidermis, endothecium, middle layers and the tapetum. The outer three wall layers perform the function of protection and help in dehiscence of anther to release the pollen. The innermost wall layer is the tapetum. It nourishes the developing pollen grains. Cells of the tapetum possess dense cytoplasm and generally have more than one nucleus.

(NCERT Ed. 2025-26, 12th, Page No. 5)

133. (4)

Viroids : In 1971, T.O. Diener discovered a new infectious agent that was smaller than viruses and caused potato spindle tuber disease. It was found to be a free RNA; it lacked the protein coat that is found in viruses, hence the name viroid. The RNA of the viroid was of low molecular weight. Since they are even smaller than viruses (which are already known to pass through such filters), viroids easily pass through bacteria-proof filters.

(NCERT Ed. 2025-26, 11th, Page No. 21)

134. (1)

Pyrenoids are proteinaceous storage bodies found in the chloroplasts of most members of Chlorophyceae (Green algae). In pteridophytes, the spores germinate to give rise to an inconspicuous, small, multicellular, free-living, mostly photosynthetic thalloid gametophyte called the prothallus. The life cycle of mosses (bryophytes) includes two gametophytic stages. The first stage is the protonema, which develops directly from a spore and is a creeping, green, branched, and often filamentous structure. The plant body in brown algae is usually attached to the substratum by a holdfast, and has a stalk, the stipe and leaf like photosynthetic organ – the frond.

(NCERT Ed. 2025-26, 11th, Page No. 26-32)

135. (3)

Micro-organisms such as *Lactobacillus* and others commonly called lactic acid bacteria (LAB) grow in milk and convert it to curd. During growth, the LAB produce acids that coagulate and partially digest the milk proteins. A small amount of curd added to the fresh milk as inoculum or starter contain millions of LAB, which at suitable temperatures multiply, thus converting milk to curd, which also improves its nutritional quality by increasing vitamin B_{12}.

(NCERT Ed. 2025-26, 12th, Page No. 151)

[Contd...

136. (1)

Prawns, cockroaches, and scorpions all belong to the Phylum Arthropoda. Members of this phylum typically possess an open circulatory system where blood (hemolymph) flows freely through body cavities rather than being confined to vessels.

(NCERT Ed. 2025-26, 11th, Page No. 44)

137. (2)

Thymidine is a nucleoside containing the sugar deoxyribose, not ribose.

DNA and RNA are polymers of nucleotides, which include a phosphate group, rather than just nucleosides.

(NCERT Ed. 2025-26, 11th, Page No. 106)

138. (2)

List-I	List-II
Acrosome	Enzymatic penetration of Ovum
Corpus Luteum	Secretes Progesterone
Leydig Cells	Synthesis of Androgens
Chorionic Villi	Forms the placenta

(NCERT Ed. 2025-26, 11th, Page No. 246)

139. (1)

In cockroaches, the hard chitinous plates (sclerites) of the exoskeleton are joined to each other by a thin, flexible articular membrane known as the arthrodial membrane. This structure allows for the necessary movement and flexibility between adjacent body segments.

(NCERT Ed. 2020-21, 11th, Page No. 111)

140. (3)

LNG-20 is a hormone-releasing intrauterine device (IUD), and implants are placed under the skin to release hormones into the bloodstream. Both methods prevent pregnancy by altering the body's hormonal balance to inhibit ovulation or affect sperm motility.

(NCERT Ed. 2025-26, 12th, Page No. 44)

141. (3)

Vital Capacity (VC) is the sum of Tidal Volume (TV), Inspiratory Reserve Volume (IRV), and Expiratory Reserve Volume (ERV). By using the formula TV = VC - (IRV + ERV), we get 4800 - (2800 + 1200) = 800 mL.

(NCERT Ed. 2025-26, 11th, Page No. 187)

142. (3)

Downstream processing involves the separation and purification of a product after the biosynthetic stage. For products like drugs, this stage must include rigorous quality control testing and clinical trials to ensure safety and efficacy before marketing.

(NCERT Ed. 2025-26, 12th, Page No. 175)

143. (4)

During the blood clotting process, an enzyme called thrombin acts on the soluble protein fibrinogen present in the plasma. Thrombin converts it into insoluble fibrin threads, which form a network to trap blood cells and create a clot.

(NCERT Ed. 2025-26, 11th, Page No. 196)

144. (4)

While sodium reabsorption occurs in multiple parts of the nephron, the majority (about 70-80%) of electrolytes, including sodium ions, are predominantly reabsorbed in the proximal convoluted tubule (PCT).

(NCERT Ed. 2025-26, 11th, Page No. 209)

145. (1)

RNA interference (RNAi) is a biological process where RNA molecules inhibit gene expression. In biotechnology, this method is specifically used to develop plants that are resistant to pests, such as nematodes, by silencing specific parasitic genes.

(NCERT Ed. 2025-26, 12th, Page No. 180)

146. (1)

Oxygen (approx. 65%) and Carbon (approx. 18.5%) are the two most abundant elements in the human body. Together, they constitute more than 80% of the total body weight, forming the backbone of almost all organic molecules.

(NCERT Ed. 2025-26, 11th, Page No. 105)

147. (4)

Growth factors are not only secreted by endocrine glands; they are also produced by various other tissues and cells throughout the body to regulate cell growth and repair.

(NCERT Ed. 2025-26, 11th, Page No. 247)

[Contd...

148. (1)

List-I	List-II
Thymus	Differentiation and maturation of T-lymphocytes
Peyer's patches	Secondary lymphoid nodules in ileum region of alimentary canal
Bone marrow	Primary site of B-lymphocyte maturation
MALT	Mucosa-associated lymphoid tissue in respiratory, digestive & urogenital tract

(NCERT Ed. 2025-26, 12th, Page No. 138)

149. (3)

The diagram shows a Graafian follicle, which contains a large antrum and a secondary oocyte. The follicle secretes high levels of estrogen; progesterone is primarily secreted by the corpus luteum after ovulation has occurred.

(NCERT Ed. 2025-26, 12th, Page No. 35)

150. (1)

Sugar gliders, Bandicoots, Wombats, and Banded anteaters (numbats) are all examples of marsupials native to Australia. Lemurs, Flying squirrels, or Wolves, which are placental mammals.

(NCERT Ed. 2025-26, 12th, Page No. 117)

151. (1)

Cardiac output is decreased when the heart rate or stroke volume drops. Parasympathetic nerve signals, primarily through the vagus nerve, release acetylcholine which slows down the heart rate, thereby reducing the overall cardiac output.

(NCERT Ed. 2025-26, 11th, Page No. 202)

152. (3)

Tobacco contains nicotine, which triggers the adrenal glands to release adrenaline and nor-adrenaline. These hormones increase heart rate and blood pressure, explaining why tobacco users have a higher risk of cardiovascular diseases.

(NCERT Ed. 2025-26, 12th, Page No. 144)

153. (2)

The study specifically focused on the history of life forms and how they have changed over time on Earth is known as evolutionary biology. Paleontology is a subset of this that focuses specifically on fossils.

(NCERT Ed. 2025-26, 12th, Page No. 110)

154. (4)

The human lungs are enclosed in a double-layered membrane called the pleural membrane. This membrane contains pleural fluid that reduces friction between the lungs and the chest wall during breathing.

(NCERT Ed. 2025-26, 11th, Page No. 184)

155. (2)

In the name *HindIII*, 'H' is the genus (*Haemophilus*), 'in' is the species (*influenzae*), 'd' is the strain (Rd), and the Roman numeral 'III' indicates the order in which the enzyme was isolated from that strain.

(NCERT Ed. 2025-26, 12th, Page No. 165)

156. (2)

Micropropagation relies on the principle of totipotency, which is the inherent ability of a single plant cell or explant to grow and regenerate into a whole new plant under specific laboratory conditions.

(NCERT Ed. 2025-26, 12th, Page No. 178)

157. (3)

In ZIFT, the zygote is transferred into the Fallopian tube, not the uterus.
IUT involves transferring embryos with more than 8 blastomeres directly into the uterus for further development.

(NCERT Ed. 2025-26, 12th, Page No. 48)

158. (2)

List-I (Common name)	List-II (Scientific name)
Sea-pen	*Pennatula*
Sea fan	*Gorgonia*
Brain coral	*Meandrina*
Portuguese Man-of-war	*Physalia*

(NCERT Ed. 2025-26, 11th, Page No. 41)

159. (2)

Exophthalmic goitre, characterized by an enlarged thyroid gland and protruding eyeballs, is a form of hyperthyroidism also known as Graves' disease.

(NCERT Ed. 2025-26, 11th, Page No. 243)

160. (4)

List-I	List-II
Micro-injection	Direct delivery of recombinant DNA into nucleus of animal cell
Biolistics	Bombardment of plant cells with high velocity particles coated with DNA
Disarmed pathogen vector	Disarmed pathogen vector
Gold or tungsten micro-particles	Used as carriers coated with DNA for plant transformation

(NCERT Ed. 2025-26, 12th, Page No. 171)

[Contd...

161. (1)

Passive immunity involves giving ready-made antibodies for a quick response.

Tight junctions stop substances from leaking across tissues; it is adhering junctions that provide cementing to neighboring cells.

(NCERT Ed. 2020-21, 11th, Page No. 102)

162. (4)

The diagram represents cartilage tissue. In this tissue, 'X' represents the chondrocytes (cartilage cells) which are enclosed in small cavities, and 'Y' represents the collagen fibers that provide structural support to the matrix.

(NCERT Ed. 2020-21, 11th, Page No. 104)

163. (3)

List-I	List-II
Sporozoite	Infective stage to humans transmitted by female *Anopheles* mosquito
Gametocyte	Sexual stage taken up by mosquito during blood meal
Primary host	Female *Anopheles* mosquito
Secondary host	Human

(NCERT Ed. 2025-26, 12th, Page No. 131)

164. (4)

Venereal diseases (STIs) are spread through sexual contact, contaminated blood, or shared needles. They are not transmitted by casual contact such as sharing clothes, utensils, or shaking hands with an infected person.

(NCERT Ed. 2025-26, 12th, Page No. 47)

165. (3)

A retroviral vector is used (not pBR322), and enzyme replacement and bone marrow transplants are not always completely curative in case of ADA deficiency.

(NCERT Ed. 2025-26, 12th, Page No. 182)

166. (3)

Cnidarians are diploblastic (having two germ layers), mostly marine animals that exhibit radial symmetry. They are characterized by the presence of cnidocytes but lack true body segmentation or a triploblastic arrangement.

(NCERT Ed. 2025-26, 11th, Page No. 41)

167. (4)

FSH, LH, and androgens (like testosterone) are all essential hormones that regulate the process of sperm production in the testes. Aldosterone is a mineralocorticoid involved in water and salt balance and has no role in spermatogenesis.

(NCERT Ed. 2025-26, 11th, Page No. 245)

168. (4)

For any DNA fragment to multiply inside a host cell, it must be linked to an 'Origin of Replication' (ori) sequence. If the alien DNA lacks this sequence, the host cell's machinery will not recognize it as a site to start replication.

(NCERT Ed. 2025-26, 12th, Page No. 164)

169. (1)

Ornithorhynchus (Platypus) is a unique mammal because it is oviparous, meaning it lays eggs. *Macropus* (Kangaroo), *Pteropus* (Flying fox), and *Elephas* (Elephant)—are all viviparous and give birth to live young.

(NCERT Ed. 2025-26, 11th, Page No. 51)

170. (4)

The formation of oogonia is completed before birth; no more oogonia are added or formed during a female's lifetime. A large number of primary follicles degenerate during the phase from birth to puberty. Therefore, at puberty only 60,000-80,000 primary follicles are left in each ovary.

(NCERT Ed. 2025-26, 12th, Page No. 32)

171. (2)

Bt cotton is genetically engineered to produce toxins that kill specific pests. To provide protection against cotton bollworms, the specific genes *cryIAc* and *cryIIAb* are introduced into the plant.

(NCERT Ed. 2025-26, 12th, Page No. 180)

172. (3)

The quaternary structure of a protein describes how two or more individual polypeptide chains (subunits) are organized and arranged in relation to each other to form a functional protein complex, such as hemoglobin.

(NCERT Ed. 2025-26, 11th, Page No. 112)

173. (3)

Chemotherapy involves the use of powerful drugs to kill rapidly dividing cancer cells. Since these drugs also affect other fast-growing healthy cells in the body, they often lead to side effects like hair loss and anemia.

(NCERT Ed. 2025-26, 12th, Page No. 142)

174. (3)

Frogs can breathe through their skin (cutaneous respiration) both in water and on land. This is possible because their skin is kept moist by mucus glands and is highly vascularized with a rich supply of blood capillaries for gas exchange.

(NCERT Ed. 2025-26, 11th, Page No. 82)

[Contd...

175. (2)

Darwin observed that although populations can grow exponentially, resources like food and space are limited. This leads to intense competition among individuals, meaning only those best adapted to the environment survive to reproduce.

(NCERT Ed. 2025-26, 12th, Page No. 119)

176. (2)

The neuromuscular junction is the specific point of contact or synapse where the terminal end of a motor neuron meets the sarcolemma of a skeletal muscle fiber to transmit electrical impulses.

(NCERT Ed. 2025-26, 11th, Page No. 222)

177. (2)

List-I	List-II
Apoenzyme	Protein part of enzyme
Competitive inhibitor	Closely resembles the substrate in its molecular structure
Coenzyme	Organic compound transiently bound to apoenzyme
Prosthetic group	Non-protein part tightly bound to enzyme

(NCERT Ed. 2025-26, 11th, Page No. 118)

178. (4)

When the body is dehydrated, the neurohypophysis (posterior pituitary) releases Vasopressin, also known as Anti-diuretic Hormone (ADH). This hormone increases water reabsorption in the kidney tubules, thereby preventing excessive urine loss (diuresis).

(NCERT Ed. 2025-26, 11th, Page No. 242)

179. (4)

In the digestive system of a frog, the bile produced by the liver and the pancreatic juice from the pancreas are both carried to the duodenum. They enter through a shared tube called the common hepato-pancreatic duct (often referred to simply as the common bile duct).

(NCERT Ed. 2025-26, 11th, Page No. 82)

180. (3)

At rest, the axonal membrane has more open potassium channels than sodium channels. The sodium-potassium pump actually transports 3 Na^+ outwards and 2 K^+ inwards to maintain the resting potential.

(NCERT Ed. 2025-26, 11th, Page No. 232)

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